

SHENG ZHAO

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I am currently involved in VR-related research: human computer interaction(HCI) & computer vision, including social computing, authentication, text entry in VR as well as 3D generation by diffusion model.

🎓 EDUCATION

Tsinghua University

August 2021 – Present

Mathematics and Physics & Electrical Engineering

Double Major

I am currently a junior with a GPA of 3.85/4 and a major GPA of 3.89/4. You can find my [transcript here](#).

- Physics of Optics&Quantum Mechanics(100), Physics of Electromagnetism(98), Calculus A1&A2(98), Mathematical Physics Equations(98), Computer Program(98).

For more information, please visit my [personal website](#).

PUBLICATIONS & MANUSCRIPTS

- **Sheng Zhao***, JunRay Zhu*, Shuning Zhang, Xueyang Wang, Fang Yi, Xin Yi: VisionSpin: A efficient Two-factor authentication in VR based on Coordinated Head and Eye Movements. (UbiComp'24, in submission)
- Xueyang Wang*, **Sheng Zhao***, Yihe Wang, Ziyu Han, Xin Ge, Xin Yi, Xin Tong: EmoBlend: Revealing the impact of facial blendshapes on emotional perception in VR. (ISMAR'24, in submission)

RESEARCH EXPERIENCES

Avatar Facial Expressions

April 2023 – December 2023

Research Assistant at Tsinghua University, Beijing, China (Advisor: [Xin Yi](#))

Keywords: HCI, social computing, blendshape, uncanny valley. We aim to investigate the impact of the intensity of facial blendshapes on user experience about emotional perception. My works include:

- Implementing VR face2face chat using TCP-based socket programming and the [Oculus SDK](#), designing UI to allow users to adjust blendshapes themselves.
- Designing both qualitative and quantitative research methods based on interviews and questionnaires.
- Leading experimenter in user research experiments and conducting data analysis.

This project is currently in the stage of paper writing, and we'll submit it to [ISMAR'24](#)(TVCG'24). Code [here](#).

VisionSpin Authentication

September 2023 – Present

Research Assistant at Tsinghua University, Beijing, China (Advisor: [Xin Yi](#))

Keywords: HCI, authentication, biometric, security. We aim to explore the feasibility and security of HMDs authentication based on the coordinated head and eye Movements. My works include:

- Project leader, proposing the current authentication framework based on drawing a 3x3 unlock pattern by eye gaze, then using the entered password pattern and behavioral biometrics for authentication.
- Using Unity to build the experiment platform, which can extract segments of eye and head movements.
- A two-factor framework of authentication by machine learning, using svm, knn, rf, mlp and resnet, to implement the classification of behavior biometrics during saccades and fixations of eye movements.

Code [here](#). We are conducting long-term experiments and the paper work, preparing to submit it to [UbiComp'24](#).

3D Generation by diffusion model

October 2023 – Present

Visiting Student of [C3I Lab](#) at Tsinghua University, Beijing, China (Advisor: [Bowen Zhou](#))

This is a research project in the field of computer vision. My works include:

- Researching the use of a pre-trained diffusion model on 2D to generate high-fidelity 3D reconstruction masks. This is particularly helpful for applications in VR and AR.
- Conducting basic experiments to validate my ideas. The project is currently in progress.

SCIENCE AND TECHNOLOGY INNOVATION COMPETITION

Tsinghua University Software Design Competition

December 2022 – January 2023

Developed a Smart Home Android app using Java, which includes features like bluetooth connectivity, weather forecast queries, and communication with hardware devices. I won the Second Prize. Code [here](#).

The 12th Tsinghua University Digital System Design Competition

August 2022 – October 2023

Designed a small car driven by MSP430 with various sensors for autonomous navigation and resource acquisition. I won the First Prize(1/60). Code can be found [here](#).

SKILLS

- Program language: C#(NET,Socket,Unity3D,VR), Python(Scikit-learn,Torch), Java(Android,Web spider)
- Machine learning & Deep learning(SVM, KNN etc. Diffusion, Transformer)
- User study design(Qualitative research, Quantitative research), Hardware design(FPGA, STM32, MSP430)
- Tools: Git, LaTeX, and Markdown.

AWARDS

<i>First Prize</i> , 36th China Physics Olympiad (CPHO), 3%	September 2020
<i>First Prize</i> , The 12th Tsinghua University Digital System Design Competition, 1%	October 2022
<i>Second Prize</i> , Tsinghua University Sports Excellence Scholarship	October 2022
<i>Second Prize</i> , Tsinghua University "Smart Coastal" Software Design Competition	January 2023
<i>First Prize</i> , Tsinghua University Scientific And Technological Innovation Scholarship	October 2023
<i>First Prize</i> , Tsinghua University Sports Excellence Scholarship	October 2023
<i>First Prize</i> , The 3rd Tsinghua Craftsman Competition (1%, Presented a representative speech)	November 2023